

Random Vectors: Expectation and Covariance Operators

A linear model is a statement about the first two moments of a random vector: its mean lies in a subspace and its covariance matrix has a given form. This chapter builds the calculus for such statements. Expectation and covariance become matrix operators that commute with linear maps, covariance matrices turn out to be exactly the nonnegative definite matrices, and a singular covariance means the vector is confined to a flat. The mean of a quadratic form needs only two moments; its variance needs four. Partitioning a vector gives cross-covariances, Schur complements and the best linear predictor, which foreshadow conditional distributions under normality.

What you need. [Chapter 1](#) (trace, spectral theorem, nonnegative definite matrices and their square roots, generalized inverses, partitioned matrices), and probability at the level of expectations, variances and independence of random variables.

2.-1.1 Roadmap

- [Mean vectors and covariance matrices](#)
- [Linear and affine transformations](#)
- [Expectation of quadratic forms](#)
- [Correlation matrices](#)
- [Moments of partitioned random vectors](#)
- [Moment generating functions and independence](#)
- [Sample moments and their computation](#)
- [Summary and notes](#)